

Detailed Summary

The project "*Serverless AWS Sentiment Microservice & Cross-Platform NLP Benchmarking*" by Kevely Diaz (May 2025) explores the design, deployment, and benchmark testing of a cloud-based sentiment analysis microservice. The service is engineered to process short text strings and categorize them into four distinct sentiment classes: positive, negative, neutral, or mixed.

Architecture & Cloud Deployment

The microservice architecture relies on Amazon Web Services (AWS) to facilitate real-time, scalable text processing. It leverages AWS Lambda for serverless computing execution and Amazon API Gateway to provision and manage incoming HTTP POST requests. The production backend uses AWS Comprehend as its native natural language processing (NLP) engine, which accepts a structured JSON payload and outputs results in a clean, reproducible framework. While the author initially intended to integrate real-time grammar feedback into the infrastructure, it was omitted due to course resource and timeline limitations.

Data Compilation & Preprocessing

To train a competitive local model, a unified database called `sentiment_data.csv` was generated by blending three distinct text channels:

1. **Sentiment140:** A standardized repository of labeled public tweets.
2. **Kaggle Dataset:** Additional categorized sentiment entries compiled by mdismielhossenabir.
3. **Synthetic Data:** Original text patterns created manually by the author to provide training representations for the "mixed" sentiment class, which was absent in the open-source corpuses.

The initial combined dataset suffered from severe class imbalance, causing early iterations of the custom model to generate heavily negative-biased classifications. To resolve this, the author implemented a rigorous data-balancing pipeline involving minority class oversampling, overrepresented class downsampling, and balanced class reweighting during training. The final balanced database was segmented using a standard 70-15-15 train-validation-test split.

Benchmarking & Performance Results

The performance of the AWS microservice was evaluated alongside three external pipelines—ChatGPT (GPT-4), Grammarly, and the custom Multinomial Logistic Regression model—using sentences extracted from a mythological passage and an excerpt from *Percy Jackson & the Olympians*.

- **ChatGPT (GPT-4):** Demonstrated the highest level of contextual nuance, standing out as the premier framework for decoding intricate, mixed emotional structures.
- **Grammarly:** Effectively mapped textual attributes but displayed a structural bias toward grammatical precision over deep emotional interpretation.
- **AWS Comprehend:** Offered fast, exceptionally reliable processing for clear, literal language, but showed contextual limitations when parsing poetic or highly figurative text blocks.

- **Custom ML Model:** Though early iterations performed poorly due to data skewing, the balanced version achieved a substantial performance recovery, accurately identifying positive, neutral, and select mixed sentiments, though it lacked the deep interpretive nuance of GPT-4.

It was concluded that the prototype successfully establishes a working link between scalable cloud deployment and NLP. Long-term future paths for the project include integrating spaCy for advanced tone analysis, enhancing classification accuracy for abstract writing, and utilizing behavioral psychology profiles to dynamically tailor writing feedback.